

RESEARCH ARTICLES

Logic of the Yeast Metabolic Cycle: Temporal Compartmentalization of Cellular Processes

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Budding yeast grown under continuous, nutrient-limited conditions exhibit robust, highly periodic cycles in the form of respiratory bursts. Microarray studies reveal that over half of the yeast genome is expressed periodically during these metabolic cycles. Genes encoding proteins having a common function exhibit similar temporal expression patterns, and genes specifying functions associated with energy and metabolism tend to be expressed with exceptionally robust periodicity. Essential cellular and metabolic events occur in synchrony with the metabolic cycle, demonstrating that key processes in a simple eukaryotic cell are compartmentalized in time.

Periodic behavior is prevalent in nature. One of the most intriguing examples of this phenomenon is circadian rhythm driven by biological clocks, found in nearly all kingdoms of life. Circadian rhythms allow organisms to coordinate their physiology with day-night cycles and may have first evolved to control cellular metabolism (1).

Similarly, the budding yeast *Saccharomyces cerevisiae* exhibits "cycles" in the form of glycolytic and respiratory oscillations (2). Such cycles were first documented over 40 years ago and can occur with a variety of period lengths both in cell-free extracts and during continuous culture (3–12). A recent study has described a ~40-min respiratory oscillation that produces a genome-wide, low-amplitude oscillation of transcription during continuous culture (10, 12). However, the molecular underpinnings responsible for controlling metabolic oscillation remain poorly understood.

We used a continuous culture system to reveal a robust, metabolic cycle in budding yeast. Here, we describe a yeast metabolic cycle (YMC) that drives the temporal, genome-wide transcription and coordination of essential cellular and metabolic processes in a manner reminiscent of the circadian cycle.

An ultradian metabolic cycle in yeast. We conducted our studies with the prototrophic, genetically tractable, diploid yeast strain CEN.PK (13). After growth to high density [optical density (OD₆₀₀) about 8 to 9] followed by a brief starvation period, the culture spontaneously began respiratory cycles as measured by oxygen consumption (Fig. 1). These highly

robust cycles were about 4 to 5 hours in length and persisted indefinitely when the cultures were continuously supplemented with low concentrations of glucose. Each cycle was characterized by a reductive, nonrespiratory phase followed by an oxidative, respiratory phase wherein the synchronized culture rapidly consumed molecular oxygen (Fig. 1).

To understand the molecular basis of these metabolic cycles, we performed microarray analysis of gene expression and assessed whether any genes were expressed periodically. Total RNA was prepared every ~25 min over three consecutive cycles (14). The high sampling rate allowed determination of the periodicities of expressed genes, including genes that are expressed only very transiently (14). The temporal expression profiles of all yeast open reading frames (ORFs) are shown in Fig. 2. By using a periodicity algorithm (14), we determined that over half of yeast genes (~3552) exhibited periodic expression patterns at a confidence level of 95% (Fig. 2C). Not surprisingly, the most common period of transcript oscillation was ~300 min (Fig. 2C),

the length of one respiratory cycle. Although transcript oscillations cycled with a period of ~300 min almost without exception, different genes were expressed maximally at entirely different times during the metabolic cycle (Fig. 2, A and B). Thus, the YMC is accompanied by a highly organized transcriptional cycle.

Genes encoding proteins associated with energy, metabolism, and protein synthesis were overrepresented in the list of periodic genes (Table 1) (14). Moreover, characterization of the periodic genes with the yeast proteome localization data (15) indicated that gene products localized to the mitochondria, cell periphery, and bud neck tended to be expressed periodically (Table 1). Of the 100 genes that exhibited the most periodic expression patterns, about two-thirds are nuclear-encoded genes involved in mitochondrial function (Table 2) (14). Taken together, these findings suggest that respiratory cycling is accompanied by cycles in metabolism and that variation in mitochondrial function is an important component of the YMC.

Cluster analysis. We turned to the most periodic genes as sentinels for the identification of clusters of genes having similar temporal expression patterns. For example, *MRPL10*, which encodes a mitochondrial ribosomal protein, is one of the most periodic genes, and its expression peaks when cells begin to cease oxygen consumption (Fig. 2B). With the use of *MRPL10* as a guide gene, we used clustering analysis to reveal a large number of genes that exhibit highly similar expression patterns to *MRPL10* (Fig. 3A and table S1) (14). Many genes within this cluster also encode components of mitochondrial ribosomes (Fig. 3A). On expanding our analysis to other annotated mitochondrial ribosomal genes, we found that 73 of 74 nuclear-encoded mitochondrial ribosomal genes displayed an extremely similar temporal expression pattern (fig. S1). The extent of coordinated expression of these genes was highest shortly after the cells ceased oxygen consumption (Fig. 3A), suggesting that

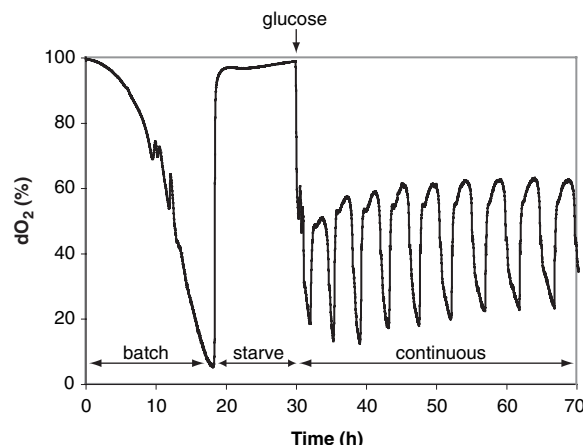
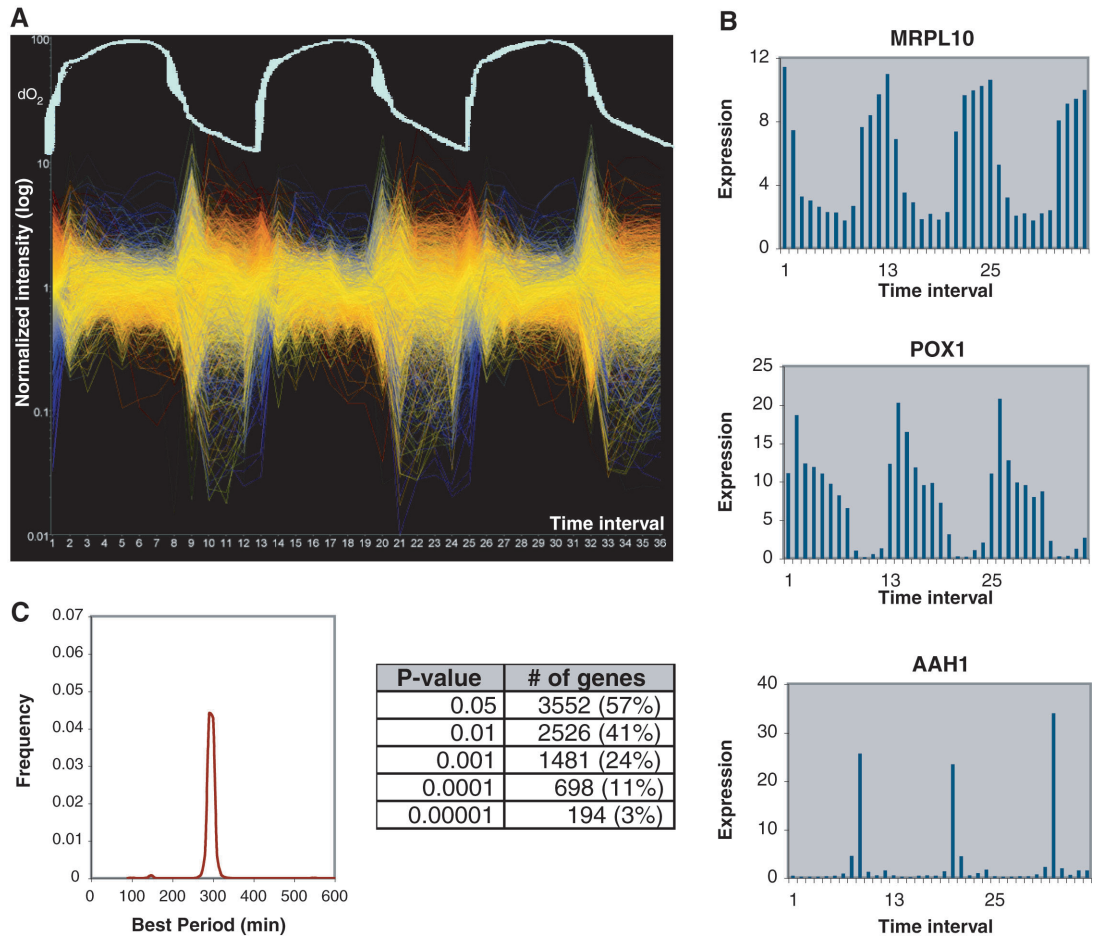


Fig. 1. The metabolic cycle of yeast. During batch mode, the cells are grown to a high density and then starved for at least 4 hours. During continuous mode (arrow), media containing glucose is introduced to the culture at a constant dilution rate (~0.09 to 0.1 hours⁻¹). dO₂ refers to dissolved oxygen concentrations (% saturation) in the media.

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Fig. 2. Periodic gene expression during the metabolic cycle. **(A)** The expression profiles of ~6209 unique expressed ORFs are displayed (three cycles, 12 time intervals per cycle, ~25 min per time interval) beneath the dO_2 trace. Dissolved oxygen increased during time intervals 1 to 7, 13 to 19, and 25 to 31 and decreased during intervals 8 to 12, 20 to 24, and 32 to 36. Spectral coloring scheme of genes has its basis in relative expression at the start of reductive phase (red indicates higher; blue, lower). **(B)** The raw expression data for three individual genes: *MRPL10* (mitochondrial ribosomal protein of the large subunit), *POX1* (fatty acyl CoA oxidase), and *AAH1* (adenine deaminase). Expression amount is in arbitrary units. **(C)** Periodicity analysis of gene expression. The most common period of transcript oscillation is ~300 min, the length of one cycle (14). The number of unique periodic genes at selected significance levels is shown (14).



cells were either rebuilding or duplicating their mitochondria at this time.

Extending the use of sentinel genes, we next turned to *POX1*, which encodes a peroxisomal fatty-acyl coenzyme A (CoA) oxidase. The peak of *POX1* gene expression occurs as dissolved oxygen accumulates during the YMC (Fig. 2B). By using *POX1* as a guide, we identified a cluster of genes with highly similar temporal expression patterns, most of which are annotated as encoding proteins involved in fatty acid oxidation and peroxisomal function (Fig. 3B and table S2). The coordinated expression of these genes strongly suggests that fatty acid oxidation preferentially occurs when the cells are not respiring.

One of the most periodic genes required for the building of cytoplasmic ribosomes is *RPL17B*, which encodes a protein component of the large (60S) ribosomal subunit. Use of *RPL17B* as a sentinel revealed a cluster of genes expressed most abundantly within a narrow window of the YMC (Fig. 3C). The vast majority of genes in this cluster also encode either ribosomal proteins or proteins involved in translation (Fig. 3C and table S3). Moreover, nearly all genes encoding cytoplasmic ribosomal proteins exhibit a similar expression pattern of peaking while the cells are respiring (fig. S2).

Table 1. Classification of periodic genes. The three most overrepresented and underrepresented functional categories (χ^2 test, P value comparing periodic with nonperiodic = 4×10^{-11}) and localizations (χ^2 test, P value = 2.2×10^{-16}) of periodic genes (14) are listed. Numbers in parentheses denote actual/expected number of genes. Functional categories according to MIPS (Munich Information Center for Protein Sequences) top-level classification.

Overrepresented		Underrepresented
<i>MIPS function</i>		
Energy (104/75)		Transcription (364/420)
Metabolism (510/460)		Protein binding (312/341)
Protein synthesis (145/123)		Protein fate (386/408)
<i>Localization</i>		
Mitochondria (217/153)		Nucleolus (44/68)
Cell periphery (54/42)		Early Golgi (10/18)
Bud neck (50/40)		Nuclear periphery (16/25)

We next performed an unbiased k -means cluster analysis of the entire microarray data set (14), which revealed three superclusters of gene expression (Fig. 3D). We termed these three superclusters Ox (oxidative), R/B (reductive/building), and R/C (reductive/charging), thereby defining three major phases of the YMC. Each of these superclusters comprises distinct subclasses of genes that are periodically expressed and peak within a certain window of the YMC (Fig. 3D).

The Ox cluster consists primarily of genes encoding ribosomal proteins, translation initiation factors, amino acid biosynthetic en-

zymes, small nuclear RNAs, RNA processing enzymes, and proteins required for the uptake and metabolism of sulfur (Fig. 3D) (14). Because protein synthesis is one of the most energy-demanding processes (16), the translation machinery may be ideally assembled when abundant amounts of adenosine triphosphate (ATP) are readily available as a consequence of an intense burst of respiration. It is particularly surprising that the Ox supercluster contains many genes that peak extremely abruptly at a single time interval in the cycle (Figs. 2 and 3C). These transcripts must be very short-lived, helping to ensure an

exceptionally tight coupling of their anabolic role with presumed access to ATP resulting from oxidative phosphorylation.

Genes of the R/B supercluster peak when cells begin to cease oxygen consumption. This supercluster consists primarily of nuclear-encoded mitochondrial genes as well as genes encoding histones, spindle pole components, and proteins required for DNA replication and cell division (Fig. 3D) (14). The majority of genes encoding mitochondrial proteins, such as those involved in mitochondrial DNA replication, respiration, and protein import, all peak during this R/B phase (14).

Lastly, genes expressed maximally during the R/C supercluster encode proteins involved in nonrespiratory modes of metabolism and protein degradation (Fig. 3D). Genes encoding components of the peroxisome, vacuole, proteasome, and ubiquitination machinery are selectively activated throughout most of the R/C phase (14).

The tight coordination of gene expression during the YMC allows prediction of regulatory motifs within gene clusters. In the *POX1*-defined peroxisomal cluster, unbiased analysis of the noncoding sequences of the top 25 clustered genes identified two potential regulatory motifs in their upstream activating sequences (UASs). Application of the MEME algorithm (17) identified 5'-WGCCGCCGW-3' (where W is A or T) and 5'-TTGGGGTAAW-3' as putative regulatory motifs at a high level of statistical significance (Fig. 4). Upon examining the mitochondrial ribosomal cluster, we observed no previously described motif in the promoter regions aside from long, A-rich sequences (Fig. 4), in which guanines were sparsely distributed within long adenine stretches. Inspection of the 3' untranslated regions (3' UTRs) of the 74 nuclear-encoded mitochondrial ribosomal genes revealed the putative regulatory sequence 5'-CNTGTANATA-3' (where N is any base) in 73 of 74 genes (Fig. 4). This motif may represent a binding site for the RNA-binding protein Puf3p (18). Out of the 74 genes queried, the only gene (*PPE1*) lacking the conserved 3' UTR motif was the one that failed to be expressed with proper temporal periodicity (fig. S1), raising the possibility that it may be misannotated (19). Further identification of conserved motifs in gene clusters may uncover previously unknown regulatory elements required for transcriptional and post-transcriptional coordination of the YMC.

The cell division cycle. We next chose as a sentinel the exceptionally periodic gene *YOX1* (Table 2), which encodes a homeodomain-containing transcriptional repressor involved in the cell cycle (20). Many genes involved in DNA replication and the cell cycle displayed highly similar temporal expression profiles, raising the possibility of a coupling between cell division and the YMC (Fig. 5A). Collectively, expression of these genes peaks in the R/B phase (Fig. 5A), suggesting that DNA replication and

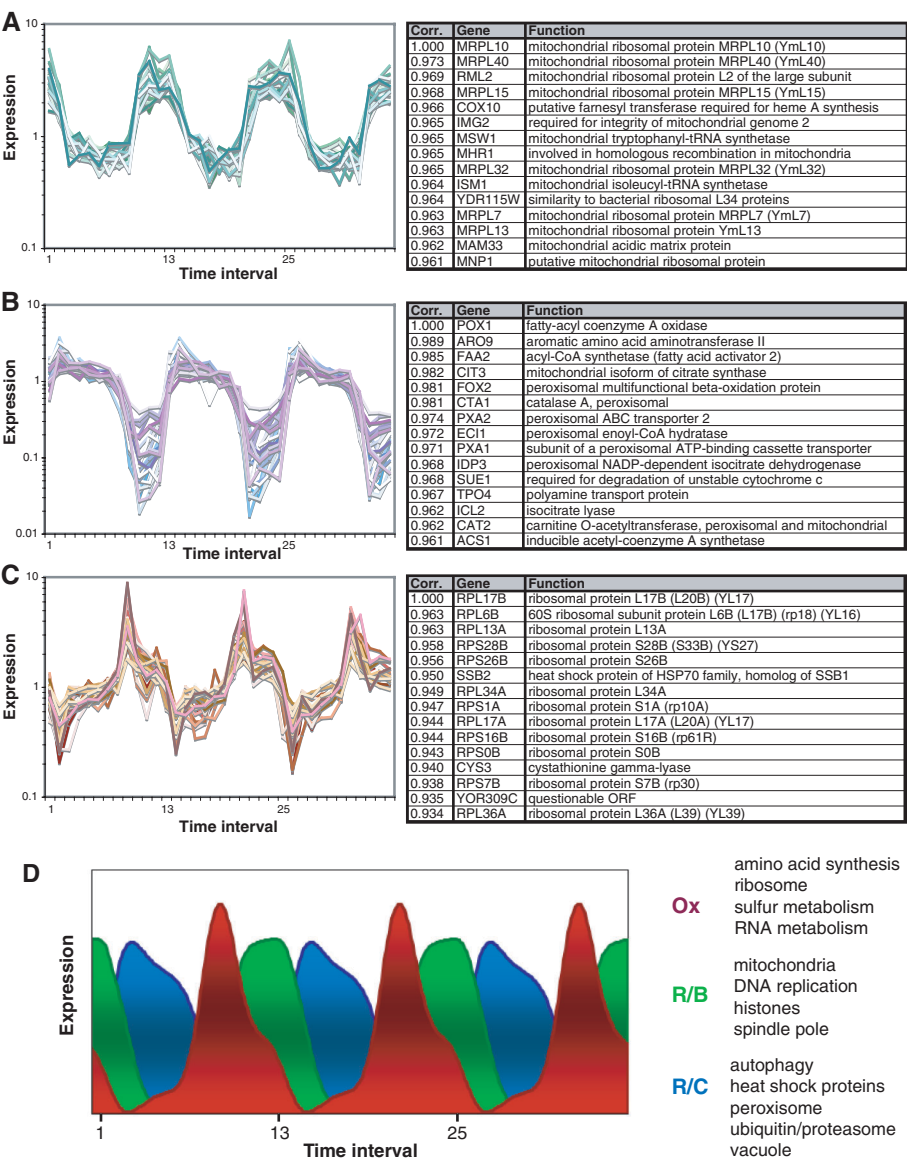


Fig. 3. Cluster analysis of gene expression during the metabolic cycle. (A to C) The mitochondrial ribosomal, peroxisomal, and ribosomal clusters. The expression profiles of the top 25 genes in each cluster are plotted together, and the identities and statistical correlations of the top 15 genes are listed to the right. Expression amount is in arbitrary units. (D) Average profiles for the three superclusters of gene expression during the metabolic cycle: Ox (oxidative), R/B (reductive, building), and R/C (reductive, charging) (14). The Ox supercluster (1023 genes) peaks roughly during time intervals 8 to 12, 20 to 24, and 32 to 36; the R/B supercluster (977 genes), during time intervals 10 to 14 and 22 to 26; and the R/C supercluster (1510 genes), during time intervals 2 to 7, 14 to 19, and 26 to 31. Examples of classes of genes in each supercluster are listed. For a complete list of genes in each supercluster, see (14).

the cell division cycle may initiate during the very late Ox phase and proceed into the R/B phase of the YMC. Cell division is strictly confined to the reductive, nonrespiratory phases (Fig. 5B). Almost no cells were observed to either replicate DNA or divide during the Ox phase (Fig. 5B). Visual examination of cells at frequent intervals showed small buds beginning to appear at the very end of the Ox phase and synchronously growing larger during progression through the R/B and R/C phases (Fig. 5B). Almost no buds were observed in the midst of Ox phase (Fig. 5B). We estimated from both fluorescence-activated cell sorting (FACS) analy-

sis and microscopic calculation of budded cells that roughly 50% of the diploid cells proceed through the cell cycle during each metabolic cycle (Fig. 5B). This value approximates the macroscopic behavior of the culture. On the basis of the dilution rate by fresh media, about half of all cells must divide each metabolic cycle to ensure stasis of culture mass. Confining the cell cycle to the reductive phases of the YMC may allow cells to minimize oxidative damage to DNA.

Dynamic changes during the metabolic cycle. The periodic transcription of genes during the YMC can be predicted to drive periodic fluctuations in metabolic output. By using ¹H

Table 2. The 40 most periodic genes, ranked by periodicity score (14).

Gene	P value	Function
MSS116	2.77×10^{-7}	Mitochondrial RNA helicase of the DEAD box family
MEF1	4.57×10^{-7}	Mitochondrial elongation factor G-like protein
BCS1	5.15×10^{-7}	Mitochondrial protein of the CDC48/PAS1/SEC18 ATPase family
MRPL10	5.42×10^{-7}	Mitochondrial ribosomal protein MRPL10 (YmL10)
ISM1	5.56×10^{-7}	Mitochondrial isoleucyl-tRNA synthetase
YLR253W	6.29×10^{-7}	Weak similarity to bacterial aminoglycoside acetyltransferase regulators
MRPL40	6.79×10^{-7}	Mitochondrial ribosomal protein MRPL40 (YmL40)
FIT2	7.40×10^{-7}	Cell wall mannoprotein
PRY2	7.45×10^{-7}	Similar to plant PR-1 class of pathogen-related proteins
COX10	8.06×10^{-7}	Farnesyl transferase required for heme A synthesis
MRPL32	8.14×10^{-7}	Mitochondrial ribosomal protein MRPL32 (YmL32)
MRPL4	8.81×10^{-7}	Mitochondrial 60S ribosomal protein L4 (YmL4)
YER163C	8.81×10^{-7}	Weak similarity to <i>Escherichia coli</i> cation transport protein
YML6	9.59×10^{-7}	Mitochondrial ribosomal protein
MSD1	1.04×10^{-6}	Mitochondrial aspartyl-tRNA synthetase
YJL213W	1.05×10^{-6}	Similarity to <i>Methanobacterium</i> arylalkylphosphatase related protein
MRPL23	1.07×10^{-6}	Mitochondrial ribosomal protein of the large subunit (YmL23)
SRL1	1.09×10^{-6}	Suppressor of rad53 lethality, cell wall mannoprotein
YDR493W	1.09×10^{-6}	Hypothetical protein
YNL300W	1.10×10^{-6}	Hypothetical protein
MRPL19	1.13×10^{-6}	Mitochondrial ribosomal protein of the large subunit (YmL19)
MTO1	1.22×10^{-6}	Mitochondrial protein
MRPL11	1.23×10^{-6}	Mitochondrial ribosomal protein MRPL11 (YmL11)
YPL103C	1.24×10^{-6}	Similarity to hypothetical <i>Mycobacterium tuberculosis</i> protein
MSW1	1.28×10^{-6}	Mitochondrial tryptophanyl-tRNA synthetase
CBP3	1.30×10^{-6}	Required for assembly of ubiquinol cytochrome-c reductase complex
SCW10	1.30×10^{-6}	Member of the glucanase gene family
YOX1	1.33×10^{-6}	Homeobox domain-containing transcriptional repressor
MRPL7	1.33×10^{-6}	Mitochondrial ribosomal protein MRPL7 (YmL7)
PET117	1.37×10^{-6}	Cytochrome c oxidase assembly factor
MRPL35	1.43×10^{-6}	Mitochondrial ribosomal protein MRPL35 (YmL35)
YJL051W	1.45×10^{-6}	Hypothetical protein
MRPL13	1.46×10^{-6}	Mitochondrial ribosomal protein YmL13
YFL052W	1.48×10^{-6}	Strong similarity to Mal63p, YPR196w, and Mal13p
CRC1	1.52×10^{-6}	Mitochondrial inner membrane carnitine transporter
RML2	1.58×10^{-6}	Mitochondrial ribosomal protein L2 of the large subunit
MSE1	1.59×10^{-6}	Mitochondrial glutamyl-tRNA synthetase
RSM19	1.62×10^{-6}	Mitochondrial ribosomal protein of the small subunit
TOS4	1.73×10^{-6}	Transcription factor, induced in G1 by SBF
MRPL31	1.75×10^{-6}	Mitochondrial ribosomal protein YmL31

Fig. 4. Prediction of regulatory motifs. Two putative motifs were found in the UAS of genes in the peroxisomal cluster (motif1 in 16 of 25 genes and motif2 in 17 of 25 genes), and one putative motif was found in both the UAS (74 of 74 genes) and the 3' UTR of nuclear-encoded mitochondrial ribosomal genes (73 of 74 genes). Pos., position.

Gene	Pos.	P-value	Motif1 (POX1-cluster UAS)	Gene	Pos.	P-value	Motif (mitochondrial ribosomal UAS)
YEL057C	-734	3.04E-07	CAGTT TAGCCGCCGA TTTTG	MRPL19	-499	6.52E-10	GCCTA GAAAGAAAAAGAAAAAGCAA AATAA
CAT2	-307	3.04E-07	TTTTT TAGCCGCCGA GACTC	RSM23	-33	3.05E-09	CAAAA CGGAAGAAAAAGAAAAAGGAAA CTCAG
SUE1	-458	3.04E-07	GTGGC TAGCCGCCGA GGTCG	MRPL17	-78	3.05E-09	ATGTA AAAAAAGAAAAAGAAAAAGAAAA CAGTT
ACS1	-226	6.07E-07	ACGAA TGCGCCGCCGA GCCTC	MRPL16	-287	8.40E-09	ATTTT GAAAAAAGAAAAAAGAAAAA AAAAA
ARO9	-233	6.07E-07	TACTA TGCGCCGCCGA CGGCC	MRP8	-996	1.68E-08	AAACC AAAAAAGAAAAAAGAAAAA GATCC
FAA2	-163	1.41E-06	ACGCC GAGCCGCCGA TGATG	RSM7	-243	2.08E-08	AAAGC AAAAAAGAAAAAAGAAAAA GATAA
PXA1	-119	2.08E-06	GAAAA TAAACGCCGA AATTA	RML2	-433	3.18E-08	ATGCA GAAAGAAAAAAGTAAAGAAA GTAAT
FOX2	-232	2.08E-06	GCAAA TAAACGCCGA GAGTT	MRP2	-49	5.75E-08	GGATT GAAAGAAAAAAGCAAGAAAA GGTA
CTA1	-249	7.72E-06	GGAAT TAGCCGCCGA AGTTG	MRPL36	-380	1.00E-07	TGATG CAGATGAAGAAAGCAAGAAA GTTTT
ARO10	-344	9.58E-06	AGGGA TAAACGCCGA TAGCC	MRP4	-954	1.20E-07	TTAAT CGAAAAAAGAAAAAAGAAAA ATTAA
PXA2	-218	9.58E-06	TTTCA TGCGCCGCCGA TAATC	RSM27	-474	1.68E-07	AGCTG GTGAAGAAAGCAAGGAAG AGGTT
YIL057C	-192	1.04E-05	ATTGC CAACGCCGCCGA AAAGG	MNP1	-887	2.74E-07	TTTTA AAGAAGAAAAAGAAAAACAATA GTATA
YCR062W	-602	1.24E-05	GATGC TGCGCCGCCGA CCTCT	MRPL7	-156	3.73E-07	AAACA GAAAGTAAAAAGCAAGGAAA AGCAT
YKL187C	-445	1.65E-05	GAATC GAAACGCCGA GCTAT	MRPL1	-387	3.73E-07	ACAAG TTGAAGAAAGAAAAAGAAAG TAGAT
POX1	-855	1.74E-05	CAGAA TACCCGCCGA TATTC	MRP49	-345	4.33E-07	ATTGA AAAAAAGAAAAATCAATAAA AGTAA

Gene	Pos.	P-value	Motif2 (POX1-cluster UAS)	Gene	Pos.	P-value	Motif (mitochondrial ribosomal 3'UTR)
TPO4	-711	1.16E-06	AATTA TGCGGCTAAA AGATC	MRPL4	26	1.98E-08	TCTCT CCTGTAAATATATA TATAT
PXA1	-138	1.16E-06	AAAAA TGCGGCTAAA AAAGG	MRPL39	77	1.65E-07	TTTTAT CCTGTAAATATATA ATGGA
ECI1	-122	1.16E-06	CGCTC TGCGGCTAAA AGAGA	MRPL13	45	2.46E-07	ATATG CCTGTACATATATA TACTC
POX1	-251	1.16E-06	AGCCG TGCGGCTAAA TCACC	MRP13	39	3.40E-07	ACATA CCTGTAAATATATA AAAGT
IDP3	-111	2.33E-06	CAATG TGCGGCTAAT TAATG	MRPL35	27	3.40E-07	AGTAT CCTGTACATATATA ATAAC
POT1	-820	3.82E-06	AATCA TGCGGCGAAA AAGAA	MRPL7	80	4.72E-07	TTCGT CATGTAAATATATA AACTG
ACS1	-297	4.56E-06	AATGA TGCGGCTCAT CCTTT	RML2	119	6.54E-07	ACAAAT CATGTAAATATATA TATTG
SUE1	-429	5.30E-06	GCCCA CTGGGCTAAA CCGCG	MRPL40	106	6.54E-07	TTCCA CCTGTATATATATA AGCAT
MDH2	-203	6.52E-06	GAGTA TGCGGCGAAA GGCAC	MRP17	39	9.08E-07	AAACA CATGTAAATATATA CATAT
FOX2	-169	7.69E-06	GGAGG ATGGGCTAAA AAAC	MRPL24	53	9.08E-07	TTTCC CCTGTATATATA AGCAT
CTA1	-559	1.05E-05	TTTTT ATGGGCTAAT CGGCT	MRPL8	25	9.08E-07	ATTCC CTTGTAATATATA GATGA
DBR1	-860	1.47E-05	CTCTA TTGGGCTAAT GAATC	IMG1	31	1.21E-06	CATCT CCTGTATATATA AAATA
YKL187C	-217	2.19E-05	CCTTT TTGGGCTAAT TACTT	MRP2	48	1.47E-06	ACGCA CTGTAAATATATA CATGA
ATO3	-92	2.19E-05	TAAAA CTGGGGGAAT ATTAG	RSM25	37	1.47E-06	ACCAA CCTGTATATATA AAAAC
YIL057C	-234	2.64E-05	TATTG TTGGGCTATT CTCTC	MRPL37	31	1.47E-06	GTATC CTTGTAATATATA TGTGT

nuclear magnetic resonance (NMR), we determined that ethanol and acetate concentrations in the media fluctuate in a robust and periodic manner during the YMC (Fig. 6A), peaking near the end of Ox phase and the beginning of R/B phase (Fig. 6A). At this time, cells have mostly finished respiration, and the onset of cell division and cellular building may favor glycolytic, fermentative metabolism. Accordingly, the expression of the gene encoding alcohol dehydrogenase *ADH1* begins to increase shortly before the rise in ethanol concentrations (14).

Many genes involved in glycolysis and the breakdown of storage carbohydrates peak in expression during the R/C phase (Fig. 6B). Furthermore, genes involved in mobilizing ethanol for the citric acid (TCA) cycle are also selectively expressed during the R/C phase of the YMC. These include *ADH2*, the alcohol dehydrogenase that converts ethanol to acetaldehyde, and *ASC1* the acetyl-CoA synthetase enzyme (Fig. 6B). The net result of these metabolic reactions is the temporally limited accumulation of acetyl-CoA (Fig. 6B). Moreover, the concurrent oxidation of fatty acids during the R/C phase should result in the production of additional acetyl-CoA units during the R/C window of the YMC (Fig. 3A). In sum, cellular metabolism during the reductive phase is predicted to be heavily devoted to the production of acetyl-CoA, preparing cells for the upcoming Ox phase. During the Ox phase, metabolism shifts to respiration as accumulated acetyl-CoA units are used for ATP production via the TCA cycle and the electron transport chain.

Metabolic output in the R/C phase would also be predicted to optimize production of NADPH (reduced form of nicotinamide adenine dinucleotide phosphate) through the induction of *GND2* and the pentose phosphate pathway (Fig. 6B). Not only is *GND2* up-regulated, but the genes encoding two enzymes (transketolase and transaldolase) that convert five-carbon sugars back to glycolytic intermediates are also coordinately activated (Fig. 6B). Both moles of NADPH made per mole of sugar are created before the return of pentose sugars to glycolysis (21). Apparently, transcriptional regulation of the genes encoding these enzymes helps tune metabolism for production of NADPH during the R/C phase. The NADPH produced is anticipated to buffer cells against oxidative stress associated with respiration. The highly concerted organization of gene expression driving the production of acetyl-CoA and NADPH allows characterization of this as the charging (R/C) phase of the yeast metabolic cycle. Lastly, ^1H NMR studies revealed almost no glucose in the media during any time of the metabolic cycle (22). Despite continuous infusion of the sugar, cells appear to immediately adsorb and metabolize available glucose.

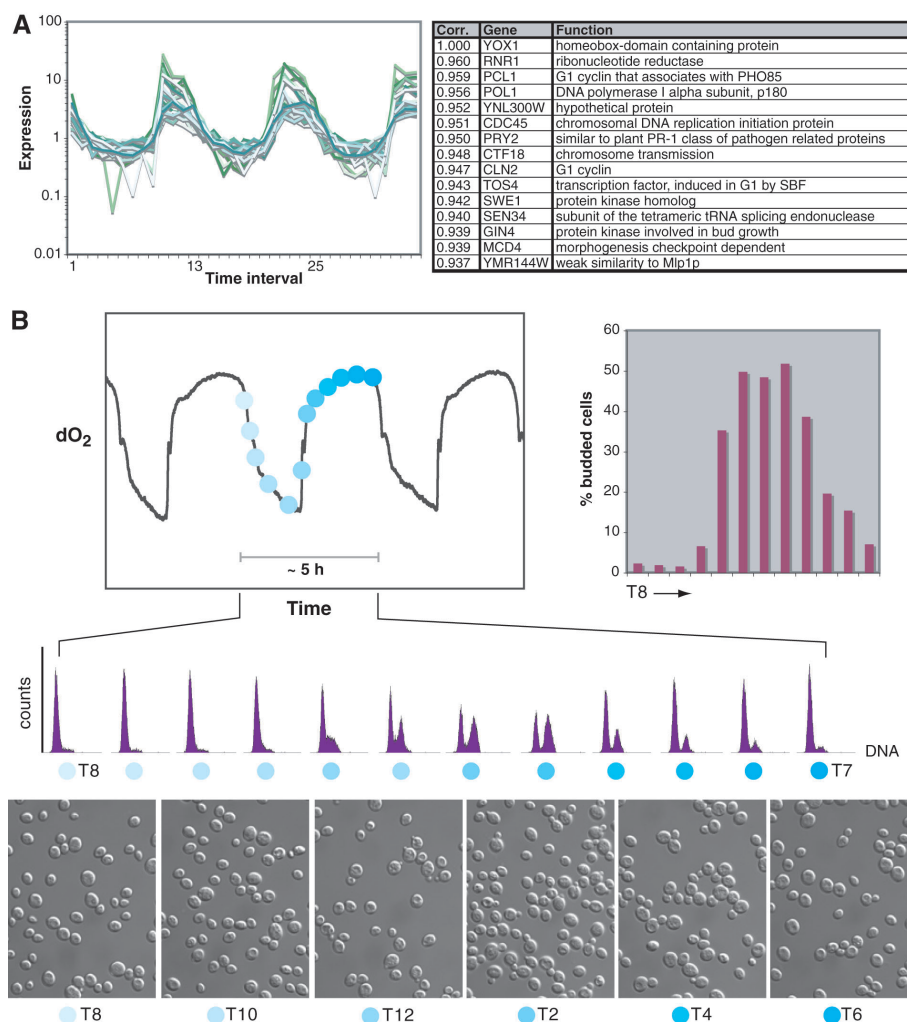


Fig. 5. Gating of the cell cycle by the metabolic cycle. (A) DNA replication and cell cycle genes cluster together. The expression profiles of the top 25 clustered genes are plotted together, and the identities and statistical correlations of the top 15 genes are listed to the right. (B) DNA replication and cell division occur only during the reductive phases of the YMC. FACS analysis of the cell population was performed at 12 time intervals over the metabolic cycle. Differential interference contrast microscopy images of the population were used to count the percentage of budded cells (minimum 300 counted per time interval, images at every other interval are shown). Counts measured by number of cells; DNA was measured by propidium iodide signal.

The large amplitude of metabolic fluctuation between oxidative and reductive phases of the YMC predicts that intracellular redox states should vary considerably as a function of the cycle. The redox-sensitive transcription factor Yap1p translocates to the nucleus upon exposure of cells to oxidative stress (23). With use of a *YAP1*-green fluorescent protein (GFP) fusion, we monitored the intracellular localization of Yap1p during the YMC. This reporter indicated nuclear localization of Yap1p during the peak of oxidative respiration (Fig. 6C). By contrast, Yap1p appeared to be exclusively cytosolic at other times of the metabolic cycle (Fig. 6C). These observations suggest that, during the Ox phase, the intracellular redox state becomes notably more oxidized, although the differences in Yap1p localization could also be due to varying concentrations of the glycolytic metabolite methylglyoxal (24). Altered redox

states within a cell may thus be a hallmark of different stages of the YMC.

We next analyzed global organelle morphology across the YMC by transmission electron microscopy (TEM) (14). The vacuole was much more prominent in Ox phase than in R/B and R/C phases of the YMC (Fig. 6D). These studies also provided evidence of autophagy in the Ox phase as seen by cellular material located within the vacuole (Fig. 6D). We then constructed a strain containing a GFP-tagged version of *VPH1*, a vacuolar ATPase subunit that resides in the vacuolar membrane (25). Visualization of these cells confirmed that vacuolar morphology changes substantially during the YMC and showed that the vacuolar membrane was more defined during the late R/C and early Ox phases (Fig. 6D). At other times, it was more dispersed and often completely undefined (Fig. 6D). This timing of vacuole emergence is con-

Fig. 6. Dynamic changes in metabolites, protein localization, and organelle morphology during the metabolic cycle. (A) Ethanol and acetate concentrations fluctuate during the metabolic cycle. Metabolites in the extracellular media were assayed by ^1H NMR (14). (B) Diagram outlining the predicted production of acetyl-CoA and NADPH during the reductive phase. Genes in blue are up-regulated notably during the R/C phase. (C) The metabolic cycle is a redox cycle. Cells expressing the redox-sensitive YAP1-GFP reporter were harvested at different times of the metabolic cycle. After fixation, Yap1p-GFP localization was visualized by fluorescence microscopy (T1, reductive; T8, oxidative) (14). (D) Dynamics of the vacuole during the metabolic cycle. (Top) TEM images of cells harvested during the reductive (red., T2) or oxidative (ox., T9) phase (n, nucleus; v, vacuole) (14). (Bottom) Images of cells expressing the vacuolar marker VPH1-GFP at different times of the cycle.

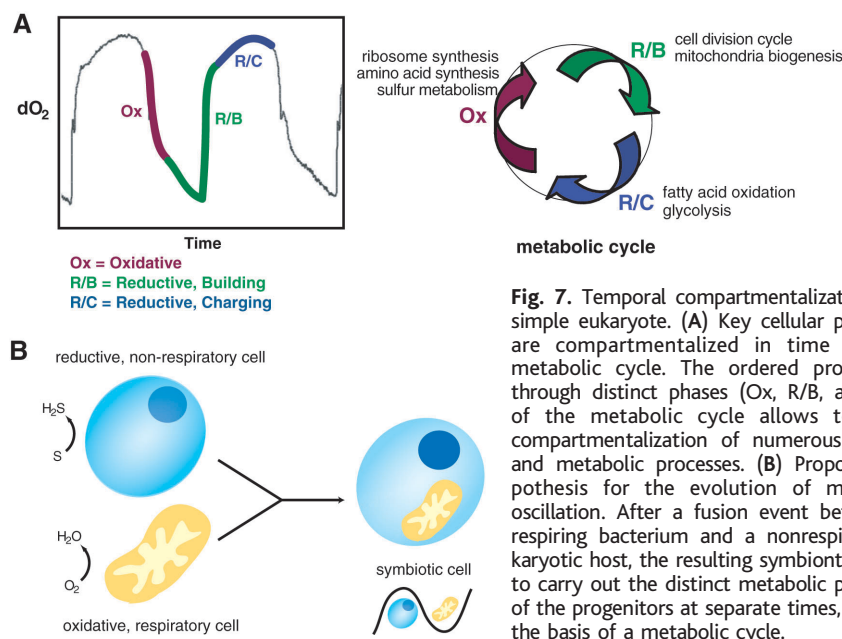
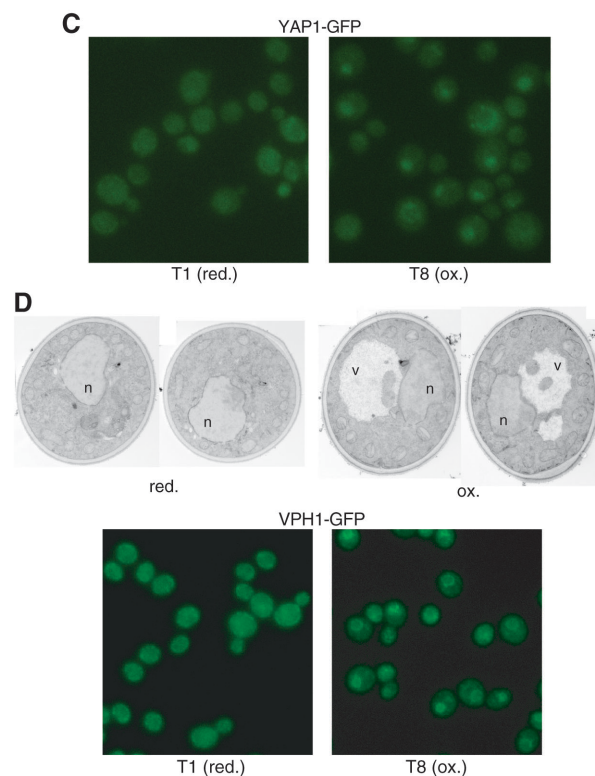
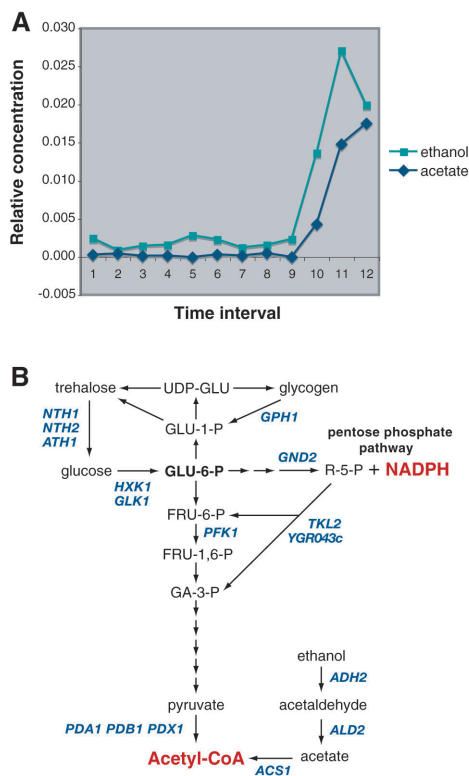


Fig. 7. Temporal compartmentalization in a simple eukaryote. (A) Key cellular processes are compartmentalized in time via the metabolic cycle. The ordered progression through distinct phases (Ox, R/B, and R/C) of the metabolic cycle allows temporal compartmentalization of numerous cellular and metabolic processes. (B) Proposed hypothesis for the evolution of metabolic oscillation. After a fusion event between a respiring bacterium and a nonrespiring eukaryotic host, the resulting symbiont evolved to carry out the distinct metabolic programs of the progenitors at separate times, forming the basis of a metabolic cycle.

sistent with results showing that genes involved in vacuolar trafficking, protein degradation, and autophagy peak during the R/C phase before respiration (Fig. 3D) (14). Thus, vacuole morphology and function are highly dynamic during the YMC. Vacuole-mediated catabolism and autophagy may be natural steps in the rebuilding of organelles and cytosol in preparation for the imminent R/B building phase.

Discussion. We have described here an unusually robust, ultradian cycle in budding yeast grown under nutrient-limited continuous conditions. Over half of the yeast genome is transcriptionally regulated in a rigidly periodic fashion as a function of this 4- to 5-hour cycle. This periodic gene expression is more robust than those seen during the circadian cycle of certain bacteria (*Synechococcus elongatus*),

fungi (*Neurospora crassa*), and almost all metazoan organisms (26–29).

The YMC controls the timing of diverse and distinct cellular and metabolic processes: Respiration, mitochondria biogenesis, ribosome biogenesis, DNA replication, cell division, fatty acid oxidation, glycolysis, and vacuole-mediated catabolism are all predicted to be precisely compartmentalized in time (Fig. 7A). Such temporal orchestration may allow cells to perform anabolic and catabolic processes in a finely coordinated and efficient fashion, helping to minimize the occurrence of futile reactions. Accordingly, genes that have functions associated with energy and metabolism tend to be expressed periodically (Table 1). Thus, cells seem to generate ATP by distinct metabolic pathways, including mitochondrial respiration, glycolysis, and fatty acid oxidation, which operate in different temporal windows of the YMC (Fig. 7A). We have focused on only a small number of the dynamic cellular changes that are temporally orchestrated throughout the YMC. Further exploration of our data set will undoubtedly reveal additional examples of events that are executed during temporally restricted windows of the YMC.

Almost all cells in our culture system rapidly become synchronized with respect to both the YMC and the cell cycle during continuous culture. This conclusion can be drawn from both macroscopic data, such as oxygen consumption, and microarray data, which show that nearly all the ~3500 cycling transcripts exhibit a periodic-

ity precisely matching the length of the YMC. It is possible that a limited number of highly specialized, secreted metabolites, analogous to those regulating quorum sensing in bacteria (30) and present at cyclically fluctuating amounts in the culture medium, control YMC synchronization. Alternatively, synchrony may be generated by fluctuating amounts of a multitude of generic metabolites, including ethanol and acetate (Fig. 6A), a cross-coupling mechanism analogous to the multinucleated, syncytial state of slime molds such as *Physarum*.

Our continuous culture system, which spontaneously generates metabolic cycles, defines a physiological paradigm that probably reflects yeast growth in the wild. The dense population of cells in the fermentor may approximate a colony exposed to nutrient-limited growth. Under these conditions, yeast cells deploy regulatory capabilities that are not typical of laboratory strains grown at log phase in rich medium. Under common laboratory growth conditions, yeast do not usually perform mitochondrial respiration. Likewise, many classes of genes, including those encoding products required for fatty acid oxidation, have minimal log-phase expression (31). By contrast, almost all of the genes required for mitochondrial and peroxisomal function are expressed and coordinately activated during a precise temporal window of the YMC.

How are the oscillating gene expression patterns of the YMC coordinated? Either messenger RNA (mRNA) synthesis, turnover, or both must combine to yield the pervasively periodic fluctuations in transcript abundance. The conserved sequences in the promoter regions of genes expressed with matching temporal kinetics that we observed indicate that regulatory periodicity will be executed, at least in part, by gene-specific transcription factors. Indeed, the mRNAs for over 60% of annotated yeast transcription factors fluctuated in abundance as a function of the YMC (14). Transcription factors that do not change in abundance might still be involved in its selective regulation. Such transcription factors may possess intrinsic metabolic sensors or be themselves coupled to signaling systems endowed with the capability to sense redox or metabolic state. The YMC will be regulated at numerous posttranscriptional levels, including selective translation and turnover of mRNA, regulated protein degradation, phosphorylation-dependent signaling, and fundamental enzyme allostery.

Our data also indicate that both DNA replication and cell division are temporally regulated as a function of the YMC. That transcriptional regulation of genes gating replication and cell division is biologically meaningful is strongly supported by FACS analysis and quantitation of budding (Fig. 5, A and B). These data demonstrate that replication and cell division are restricted exclusively to the reductive phases of the YMC. Cells may have

evolved this tight coupling to ensure that the cell cycle evades the potentially mutagenic redox environment of the oxidative respiratory phase. As shown in Fig. 7A, there is a conceptual relation between the tripartite Ox, R/B, and R/C oscillation of the metabolic cycle and the phases of the cell division cycle (G1, S, G2, and M). Such gating of cell division has been observed as a function of circadian or ultradian rhythm in other species (11, 32–34).

What are the evolutionary origins of the YMC? About two-thirds of the most periodic transcripts in the YMC encode components of mitochondria (Table 2) (12). Thus, the timing of construction of new mitochondria, or reconstruction of spent mitochondria, is probably of exceptional importance to the YMC. These observations, coupled with the fact that mitochondrial respiration is restricted to a brief window during the YMC, highlight the ability of yeast cells to temporally regulate oxidative phosphorylation. Might it be that an ancestral symbiont (33), endowed with the capacity to generate ATP by both respiratory and reductive pathways, used the two pathways in an oscillatory manner (Fig. 7B)? Given the expectation that the two pathways were optimally tuned in the progenitors of the symbiont, we speculate that the resulting hybrid found a way of leaving each pathway close to its own optimal, physiological milieu by temporal compartmentalization.

Temporal compartmentalization of metabolic function also appears to take place during the circadian cycle of flies and mice (25, 26). The primitive cyanobacterium *Synechococcus elongatus*, which conducts both photosynthesis and nitrogen fixation, uses its circadian regulatory apparatus to ensure that these biochemically incompatible pathways are executed at temporally distinct phases of the circadian cycle (34). The circadian cycle drives the periodic expression of many genes encoding the rate-limiting enzymes of numerous metabolic processes (25, 26). Restricted feeding can entrain the circadian cycle (35, 36), perhaps through metabolic feedback impinging directly on the transcription factors that themselves regulate circadian rhythm (37, 38). Metabolic oscillation may, therefore, constitute the primordial device upon which the divergent circadian and ultradian biological oscillators of modern organisms have been built (1).

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Supporting Online Material

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Materials and Methods
Figs S1 and S2
Tables S1 to S3
References

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ERRATUM

Post date 17 February 2006

Research Articles: "Logic of the yeast metabolic cycle: temporal compartmentalization of cellular processes" by B. P. Tu *et al.* (18 Nov. 2005, p. 1152). Several references were cited in the wrong places in the text. The correct sentences and citations are as follows: "A recent study described a ~40-min respiratory oscillation that produces a genome-wide, low-amplitude oscillation of transcription during continuous culture (11, 12)... About two-thirds of the most periodic transcripts in the YMC encode components of mitochondria (Table 2) (14)... Might it be that an ancestral symbiont (35), endowed with the capacity to generate ATP by both respiratory and reductive pathways, used the two pathways in an oscillatory manner (Fig. 7B)?... Temporal compartmentalization of metabolic function also appears to take place during the circadian cycle of flies and mice (27, 28). The primitive cyanobacterium *Synechococcus elongatus*, which conducts both photosynthesis and nitrogen fixation, uses its circadian regulatory apparatus to ensure that these biochemically incompatible pathways are executed at temporally distinct phases of the circadian cycle (36). The circadian cycle drives the periodic expression of many genes encoding the rate-limiting enzymes of numerous metabolic processes (27, 28). Restricted feeding can entrain the circadian cycle (35, 36), perhaps through metabolic feedback impinging directly on the transcription factors that themselves regulate circadian rhythm (39, 40)."



Logic of the Yeast Metabolic Cycle: Temporal Compartmentalization of Cellular Processes

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